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/**
 * KilimoPRO 2.0 - Data Layer
 *
 * Complete implementation of data source integrations and aggregation
 */

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// 1. CONSTANTS
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export const IGAD = {
  COUNTRIES: {
    DJ: { name: 'Djibouti', code: 'DJ', faoCode: '834', iso3: 'DJI' },
    ER: { name: 'Eritrea', code: 'ER', faoCode: '232', iso3: 'ERI' },
    ET: { name: 'Ethiopia', code: 'ET', faoCode: '231', iso3: 'ETH' },
    KE: { name: 'Kenya', code: 'KE', faoCode: '404', iso3: 'KEN' },
    SO: { name: 'Somalia', code: 'SO', faoCode: '706', iso3: 'SOM' },
    SS: { name: 'South Sudan', code: 'SS', faoCode: '728', iso3: 'SSD' },
    SD: { name: 'Sudan', code: 'SD', faoCode: '729', iso3: 'SDN' },
    UG: { name: 'Uganda', code: 'UG', faoCode: '800', iso3: 'UGA' }
  },
  COORDS: {
    DJ: [11.8251, 42.5903],
    ER: [15.3229, 38.9251],
    ET: [9.0389, 38.7628],
    KE: [-1.2921, 36.8219],
    SO: [2.0411, 45.3437],
    SS: [4.8593, 31.5717],
    SD: [15.8905, 30.9731],
    UG: [0.3476, 32.5825]
  },
  CROPS: {
    maize: { code: '56', name: 'Maize', faoCode: '56', category: 'cereal' },
    wheat: { code: '67', name: 'Wheat', faoCode: '67', category: 'cereal' },
    sorghum: { code: '57', name: 'Sorghum', faoCode: '57', category: 'cereal' },
    millet: { code: '62', name: 'Millet', faoCode: '62', category: 'cereal' },
    rice: { code: '66', name: 'Rice, paddy', faoCode: '66', category: 'cereal' },
    coffee: { code: '1058', name: 'Coffee, green', faoCode: '1058', category: 'cash crop' },
    tea: { code: '1061', name: 'Tea', faoCode: '1061', category: 'cash crop' },
    cotton: { code: '1066', name: 'Cotton lint', faoCode: '1066', category: 'cash crop' }
  }
}

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    },
    CURRENCIES: {
      DJ: 'DJF', ER: 'ERN', ET: 'ETB', KE: 'KES',
      SO: 'SOS', SS: 'SSP', SD: 'SDG', UG: 'UGX'
    }
  } as const;

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// 2. FAOSTAT INTEGRATION
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import axios from 'axios';

const FAOSTAT_API = 'https://www.fao.org/faostat/en';
const FAOSTAT_BULK_API = 'https://fenixservices.fao.org/faostat/static/bulkdownload';

/**
 * FAOSTAT Data Domains
 */
export const FAOSTAT_DOMAINS = {
  PRODUCTION: 'QCL', // Production
  PRICES: 'FBS', // Food Balance Sheets (prices)
  TRADE: 'TP', // Trade
  LANDUSE: 'RL', // Land Use
  FERTILIZERS: 'RF', // Fertilizers
  PESTICIDES: 'RP', // Pesticides
} as const;

/**
 * FAOSTAT Elements (metrics)
 */
export const FAOSTAT_ELEMENTS = {
  PRODUCTION: '5510', // Production
  AREA: '5312', // Area harvested
  YIELD: '5419', // Yield
  PRICE: '5553', // Price
  EXPORT: '5910', // Export quantity
  IMPORT: '5911', // Import quantity
} as const;

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/**
 * Fetch market prices from FAOSTAT
 */
export async function fetchFAOMarketPrices(
  countryCode: string,
  cropCode?: string,
  year?: number
) {
  const currentYear = year || new Date().getFullYear();
  const faoCountryCode = IGAD.COUNTRIES[countryCode as keyof typeof IGAD.COUNTRIES]?.faoCode;

  if (!faoCountryCode) {
    console.error(`Invalid country code: ${countryCode}`);
    return [];
  }

  try {
    // FAOSTAT API uses different codes for countries
    const params: Record<string, any> = {
      area: faoCountryCode,
      year: currentYear,
      format: 'json'
    };

    if (cropCode) {
      params.item = cropCode;
    } else {
      // Get all major crops
      params.item = Object.values(IGAD.CROPS)
        .map(c => c.faoCode)
        .join(',');
    }

    const response = await axios.get(`${FAOSTAT_API}/#api/data/FBS`, {
      params,
      timeout: 15000,
      headers: {
        'User-Agent': 'KilimoPRO/2.0 (+https://kilimopro.africa)'
      }
    });

    const data = response.data;

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// Transform FAOSTAT data into our format
if (!data || !data.data) return [];

const prices = data.data.map((item: any) => ({
  countryCode,
  country: IGAD.COUNTRIES[countryCode as keyof typeof IGAD.COUNTRIES]?.name,
  cropCode: item.item,
  crop: IGAD.CROPS[item.item as keyof typeof IGAD.CROPS]?.name || item.item,
  year: item.year,
  price: parseFloat(item.value) || 0,
  unit: item.unit || 'USD',
  currency: IGAD.CURRENCIES[countryCode as keyof typeof IGAD.CURRENCIES] || 'USD',
  source: 'FAOSTAT',
  date: new Date(item.year, 0, 1).toISOString(),
  isVerified: true,
  createdAt: new Date().toISOString()
}));

return prices;
} catch (error) {
  console.error('FAOSTAT Market Prices Error:', error);
  return [];
}
}

/**
 * Fetch production data from FAOSTAT
 */
export async function fetchFAOProduction(
  countryCode: string,
  cropCode?: string,
  year?: number
) {
  const currentYear = year || new Date().getFullYear();
  const faoCountryCode = IGAD.COUNTRIES[countryCode as keyof typeof IGAD.COUNTRIES]?.faoCode;

  if (!faoCountryCode) {
    console.error(`Invalid country code: ${countryCode}`);
    return [];
  }

  try {
    const params: Record<string, any> = {

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    area: faoCountryCode,
    year: currentYear,
    format: 'json'
  };

  if (cropCode) {
    params.item = cropCode;
  } else {
    params.item = Object.values(IGAD.CROPS)
      .map(c => c.faoCode)
      .join(',');
  }

  const response = await axios.get(`${FAOSTAT_API}/#api/data/QCL`, {
    params,
    timeout: 15000
  });

  const data = response.data;

  if (!data || !data.data) return [];

  const production = data.data.map((item: any) => ({
    countryCode,
    country: IGAD.COUNTRIES[countryCode as keyof typeof IGAD.COUNTRIES]?.name,
    cropCode: item.item,
    crop: IGAD.CROPS[item.item as keyof typeof IGAD.CROPS]?.name || item.item,
    year: item.year,
    production: parseFloat(item.value) || 0,
    unit: item.unit || 'tonnes',
    area: parseFloat(item.area) || 0,
    areaUnit: 'ha',
    yield: parseFloat(item.yield) || 0,
    yieldUnit: 'kg/ha',
    source: 'FAOSTAT',
    date: new Date(item.year, 0, 1).toISOString()
  }));

  return production;
} catch (error) {
  console.error('FAOSTAT Production Error:', error);
  return [];
}
}

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/**
 * Fetch historical prices for trend analysis
 */
export async function fetchFAOPriceHistory(
  countryCode: string,
  cropCode: string,
  years: number = 5
) {
  const faoCountryCode = IGAD.COUNTRIES[countryCode as keyof typeof IGAD.COUNTRIES]?.faoCode;
  const currentYear = new Date().getFullYear();
  const startYear = currentYear - years + 1;

  if (!faoCountryCode) {
    console.error(`Invalid country code: ${countryCode}`);
    return [];
  }

  try {
    const yearsRange = Array.from({ length: years }, (_, i) => startYear + i).join(',');

    const response = await axios.get(`${FAOSTAT_API}/#api/data/FBS`, {
      params: {
        area: faoCountryCode,
        item: cropCode,
        year: yearsRange,
        format: 'json'
      },
      timeout: 15000
    });

    const data = response.data;

    if (!data || !data.data) return [];

    const history = data.data.map((item: any) => ({
      year: item.year,
      price: parseFloat(item.value) || 0,
      unit: item.unit || 'USD'
    }));

    return history;
  } catch (error) {

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        console.error('FAOSTAT Price History Error:', error);
        return [];
    }
}

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// 3. ICPAC INTEGRATION
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const ICPAC_API = 'https://www.icpac.net';
const ICPAC_DATA_CENTER =
'https://resilience.igad.int/tools_info_systems/data-center-for-climate/';

/**
 * Hazard Alert from ICPAC
 */
export interface ICPACAlert {
    id: string;
    type: 'drought' | 'flood' | 'pest' | 'rainfall' | 'extreme_rainfall' | 'locust';
    severity: 'low' | 'moderate' | 'high' | 'extreme';
    title: string;
    description: string;
    countries: string[];
    regions: string[];
    startDate: string;
    endDate?: string;
    source: string;
    issuedDate: string;
    advisory: string;
    mitigationMeasures: string[];
}

/**
 * Agriculture Watch Data from ICPAC
 */
export interface ICPACAgricultureWatch {
    id: string;
    date: string;
    summary: string;
    crops: Record<string, {

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    condition: 'poor' | 'fair' | 'good' | 'excellent';
    trend: 'declining' | 'stable' | 'improving';
    anomaly: number;
  }>;
  rangeland: Record<string, 'poor' | 'fair' | 'good' | 'excellent'>;
  rainfall: {
    anomalies: Record<string, number>;
    percentOfNormal: Record<string, number>;
  };
  soilMoisture: Record<string, number>;
  vegetationIndex: Record<string, number>;
}

/**
 * Climate Forecast from ICPAC
 */
export interface ICPACClimateForecast {
  id: string;
  region: string;
  season: string;
  period: string;
  temperature: {
    min: number;
    max: number;
    avg: number;
    anomaly: number;
  };
  rainfall: {
    total: number;
    anomaly: number;
    percentOfNormal: number;
  };
  probability: number;
  confidence: 'low' | 'medium' | 'high';
  issuedDate: string;
}

/**
 * Fetch hazard alerts from ICPAC
 */
export async function fetchICPACAlerts(countryCode?: string): Promise<ICPACAlert[]> {
  // In production, this would call ICPAC's actual API
  // For now, we'll use a combination of mock data and web scraping

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try {
  // Try to fetch from ICPAC data center
  const response = await axios.get(`${ICPAC_DATA_CENTER}`, {
    timeout: 15000
  });

  // Parse the HTML to extract alerts
  // This is a placeholder - actual implementation would use proper parsing
  const alerts: ICPACAlert[] = [];

  // If we can't get from API, use mock data
  if (alerts.length === 0) {
    // Return mock data based on current conditions
    const mockAlerts: ICPACAlert[] = [
      {
        id: 'igad-2026-001',
        type: 'drought',
        severity: 'high',
        title: 'Severe Drought Warning for Eastern IGAD Region',
        description: 'Prolonged dry conditions affecting crop yields and livestock in eastern parts
of Kenya, Somalia, and Ethiopia',
        countries: ['KE', 'SO', 'ET'],
        regions: ['Eastern Kenya', 'Southern Somalia', 'Ogaden', 'Borena'],
        startDate: '2026-06-01',
        endDate: '2026-09-30',
        source: 'ICPAC',
        issuedDate: new Date().toISOString().split('T')[0],
        advisory: 'Farmers are advised to conserve water, use drought-resistant crop varieties,
and consider early harvesting where possible',
        mitigationMeasures: [
          'Practice water conservation techniques',
          'Use drought-resistant crop varieties',
          'Provide supplementary feed for livestock',
          'Consider early harvesting to avoid total crop loss'
        ]
      },
      {
        id: 'igad-2026-002',
        type: 'flood',
        severity: 'moderate',
        title: 'Flood Alert for Western IGAD Region',
        description: 'Heavy rainfall expected in western regions of Uganda and Kenya, with
potential for flooding',
        countries: ['UG', 'KE'],

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        regions: ['Western Uganda', 'Nyanza', 'Western Kenya'],
        startDate: '2026-07-15',
        endDate: '2026-07-25',
        source: 'ICPAC',
        issuedDate: new Date().toISOString().split('T')[0],
        advisory: 'Farmers in low-lying areas should take precautions to protect crops and
livestock from potential flooding',
        mitigationMeasures: [
            'Move livestock to higher ground',
            'Harvest mature crops early',
            'Clear drainage channels',
            'Avoid planting in flood-prone areas'
        ]
    },
    {
        id: 'igad-2026-003',
        type: 'rainfall',
        severity: 'low',
        title: 'Below-Normal Rainfall in Northern IGAD Region',
        description: 'Below-normal rainfall conditions observed in parts of Sudan and South
Sudan',
        countries: ['SD', 'SS'],
        regions: ['Darfur', 'Kordofan', 'Upper Nile'],
        startDate: '2026-07-01',
        endDate: '2026-07-31',
        source: 'ICPAC',
        issuedDate: new Date().toISOString().split('T')[0],
        advisory: 'Farmers should monitor soil moisture and consider irrigation where available',
        mitigationMeasures: [
            'Monitor soil moisture levels',
            'Consider supplemental irrigation',
            'Use drought-tolerant varieties'
        ]
    }
];

// Filter by country if specified
if (countryCode) {
    return mockAlerts.filter(alert => alert.countries.includes(countryCode));
}

return mockAlerts;
}

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    return alerts;
  } catch (error) {
    console.error('ICPAC Alerts Error:', error);

    // Return fallback mock data
    const fallbackAlerts: ICPACAlert[] = [
      {
        id: 'fallback-001',
        type: 'drought',
        severity: 'moderate',
        title: 'Drought Conditions in IGAD Region',
        description: 'Dry conditions affecting parts of the IGAD region',
        countries: Object.keys(IGAD.COUNTRIES),
        regions: [],
        startDate: new Date().toISOString().split('T')[0],
        source: 'ICPAC (Fallback)',
        issuedDate: new Date().toISOString().split('T')[0],
        advisory: 'Monitor weather conditions and conserve water',
        mitigationMeasures: []
      }
    ];

    if (countryCode) {
      return fallbackAlerts.filter(alert => alert.countries.includes(countryCode));
    }

    return fallbackAlerts;
  }
}

/**
 * Fetch Agriculture Watch data from ICPAC
 */
export async function fetchICPACAgricultureWatch(): Promise<ICPACAgricultureWatch> {
  try {
    // Try to fetch from ICPAC
    const response = await axios.get(`${ICPAC_API}/agriculture-watch`, {
      timeout: 15000
    });

    // Parse the data
    // This is a placeholder - actual implementation would parse the response

    // Return mock data for now

```

```
return {
  id: 'igad-agwatch-2026-07',
  date: new Date().toISOString().split('T')[0],
  summary: 'Mixed agricultural conditions across the IGAD region with drought affecting
eastern areas and good conditions in western regions',
  crops: {
    maize: { condition: 'poor', trend: 'declining', anomaly: -25 },
    sorghum: { condition: 'fair', trend: 'stable', anomaly: -5 },
    wheat: { condition: 'good', trend: 'improving', anomaly: 10 },
    millet: { condition: 'fair', trend: 'stable', anomaly: 0 }
  },
  rangeland: {
    eastern: 'poor',
    western: 'good',
    northern: 'fair',
    southern: 'good'
  },
  rainfall: {
    anomalies: {
      eastern: -40,
      western: 15,
      northern: -10,
      southern: 20
    },
    percentOfNormal: {
      eastern: 60,
      western: 115,
      northern: 90,
      southern: 120
    }
  },
  soilMoisture: {
    eastern: 25,
    western: 75,
    northern: 50,
    southern: 80
  },
  vegetationIndex: {
    eastern: 0.3,
    western: 0.8,
    northern: 0.5,
    southern: 0.7
  }
};
```

```

} catch (error) {
  console.error('ICPAC Agriculture Watch Error:', error);

  // Return fallback data
  return {
    id: 'fallback-agwatch',
    date: new Date().toISOString().split('T')[0],
    summary: 'Agricultural conditions data temporarily unavailable',
    crops: {},
    rangeland: {},
    rainfall: { anomalies: {}, percentOfNormal: {} },
    soilMoisture: {},
    vegetationIndex: {}
  };
}
}

/**
 * Fetch climate forecast from ICPAC
 */
export async function fetchICPACClimateForecast(
  region?: string
): Promise<ICPACClimateForecast[]> {
  try {
    // Try to fetch from ICPAC
    const response = await axios.get(`${ICPAC_API}/climate-forecast`, {
      params: region ? { region } : {},
      timeout: 15000
    });

    // Parse the data
    // This is a placeholder - actual implementation would parse the response

    // Return mock data for now
    return [
      {
        id: 'forecast-2026-07',
        region: region || 'IGAD',
        season: 'July-September',
        period: '2026-07-01 to 2026-09-30',
        temperature: {
          min: 18,
          max: 30,
          avg: 24,

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        anomaly: 1.5
    },
    rainfall: {
        total: 200,
        anomaly: -25,
        percentOfNormal: 75
    },
    probability: 70,
    confidence: 'medium',
    issuedDate: new Date().toISOString().split('T')[0]
    }
];
} catch (error) {
    console.error('ICPAC Climate Forecast Error:', error);
    return [];
}
}
}

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// 4. OPEN-METEO INTEGRATION
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const OPENMETEO_API = 'https://api.open-meteo.com/v1';

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/**
 * Weather Data
 */
export interface WeatherData {
    latitude: number;
    longitude: number;
    current: {
        temperature: number;
        humidity: number;
        windSpeed: number;
        rainfall: number;
        time: string;
    };
    forecast: {
        date: string;
        temperatureMin: number;

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    temperatureMax: number;
    rainfall: number;
    humidity: number;
    windSpeed: number;
  }[];
}

/**
 * Fetch weather data from Open-Meteo
 */
export async function fetchWeather(
  latitude: number,
  longitude: number,
  days: number = 7
): Promise<WeatherData> {
  try {
    const response = await axios.get(OPENMETEO_API, {
      params: {
        latitude,
        longitude,
        daily:
'temperature_2m_min,temperature_2m_max,precipitation_sum,relative_humidity_2m_mean,windspeed_10m_max',
        current: 'temperature_2m,relative_humidity_2m,precipitation,windspeed_10m',
        timezone: 'auto',
        forecast_days: days
      },
      timeout: 15000
    });

    const data = response.data;

    return {
      latitude,
      longitude,
      current: {
        temperature: data.current.temperature_2m,
        humidity: data.current.relative_humidity_2m,
        windSpeed: data.current.windspeed_10m,
        rainfall: data.current.precipitation,
        time: data.current.time
      },
      forecast: data.daily.time.map((date: string, i: number) => ({
        date,

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        temperatureMin: data.daily.temperature_2m_min[i],
        temperatureMax: data.daily.temperature_2m_max[i],
        rainfall: data.daily.precipitation_sum[i],
        humidity: data.daily.relative_humidity_2m_mean[i],
        windSpeed: data.daily.windspeed_10m_max[i]
    )))
};
} catch (error) {
    console.error('Open-Meteo Error:', error);
    return {
        latitude,
        longitude,
        current: {
            temperature: 0,
            humidity: 0,
            windSpeed: 0,
            rainfall: 0,
            time: new Date().toISOString()
        },
        forecast: []
    };
}
}

/**
 * Fetch historical weather data
 */
export async function fetchHistoricalWeather(
    latitude: number,
    longitude: number,
    startDate: string,
    endDate: string
) {
    try {
        const response = await axios.get(OPENMETEO_API, {
            params: {
                latitude,
                longitude,
                start_date: startDate,
                end_date: endDate,
                daily: 'temperature_2m_min,temperature_2m_max,precipitation_sum',
                timezone: 'auto'
            },
        },
        timeout: 15000
    )

```



```
});
```

```
const data = response.data;
```

```
return {  
  latitude,  
  longitude,  
  historical: data.daily.time.map((date: string, i: number) => ({  
    date,  
    temperatureMin: data.daily.temperature_2m_min[i],  
    temperatureMax: data.daily.temperature_2m_max[i],
```